

AutoGuardian Anti-Theft System - Hardware BOM

1. ESP32-WROOM-32 DEVKIT V1

- Main control unit (Wi-Fi + GSM + sensor management).

2. SIM800L GSM GPRS Module

- Sends SMS alerts or HTTP requests via GSM when Wi-Fi is unavailable.

3. NEO-6M GPS Module with Antenna

- Provides real-time car GPS coordinates.

4. 1-Channel 5V Relay Module (Opto-isolated)

- Acts as a kill switch to disable ignition or fuel relay.

5. Active Buzzer Module 5V

- Audible alarm during alerts or arming/disarming.

6. TTP223 Capacitive Touch Sensor Module

- Touch input for arming/disarming the system.

7. MPU-6050 Accelerometer + Gyroscope

- Detects motion, tilt, or shock to identify tampering.

8. DC-DC Buck Converter (e.g., LM2596)

- Steps down car 12V to 5V or 4V for modules.

9. AMS1117-4.0V Regulator

- Supplies stable 4.0V for SIM800L module.

10. 4-channel Logic Level Shifter

- Converts 5V signals to 3.3V safely (optional).

11. SSD1306 128x64 I2C OLED Display

- Displays system status and time (optional).

12. Jumper Wires + Breadboard

- Used for prototyping only. In final build, use soldered connections or PCBs.

13. ABS Project Box Enclosure

- Protective casing for installation in the car.

14. 3.7V Li-ion Battery + TP4056

- Backup power in case main car battery is cut off (optional).